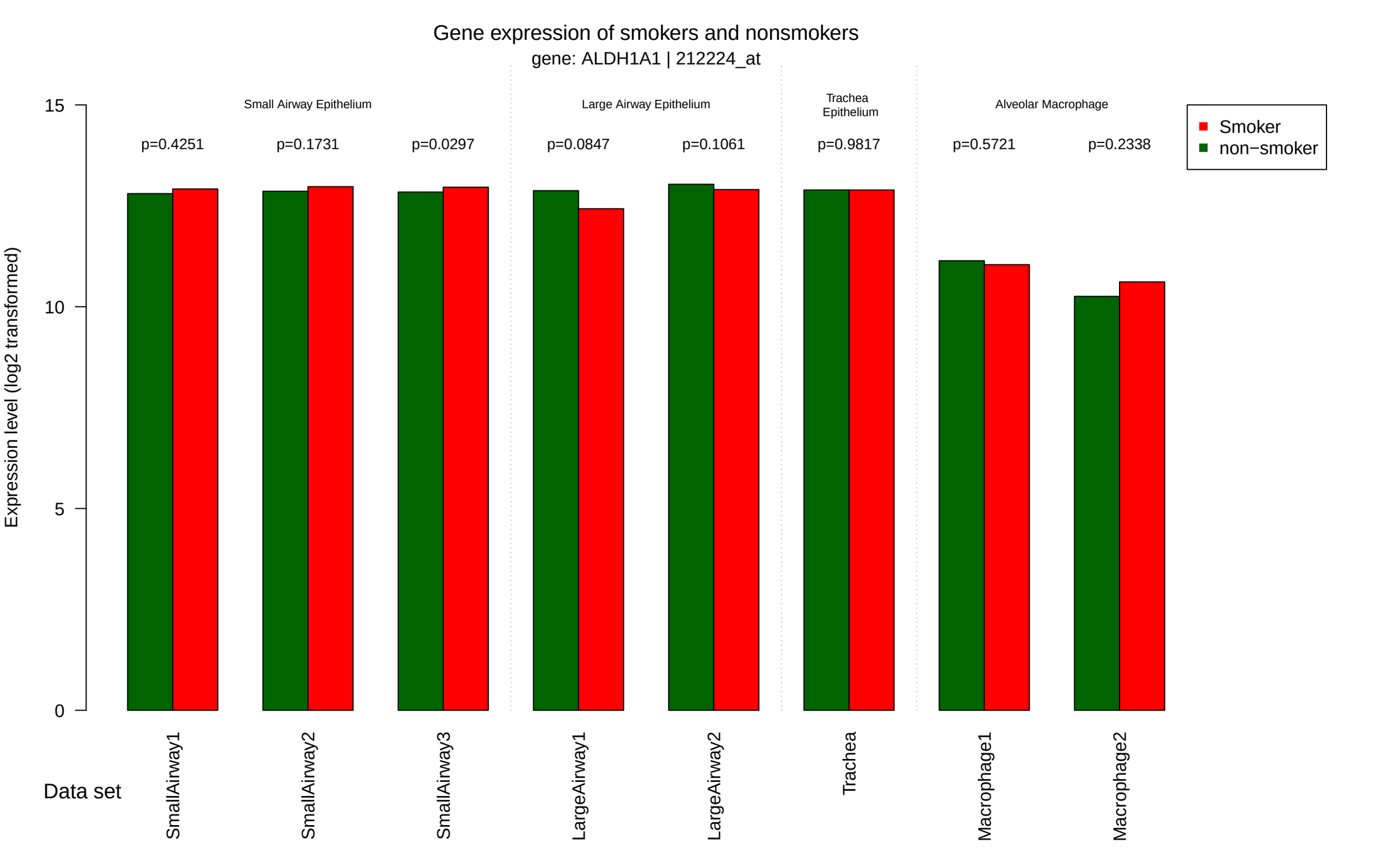
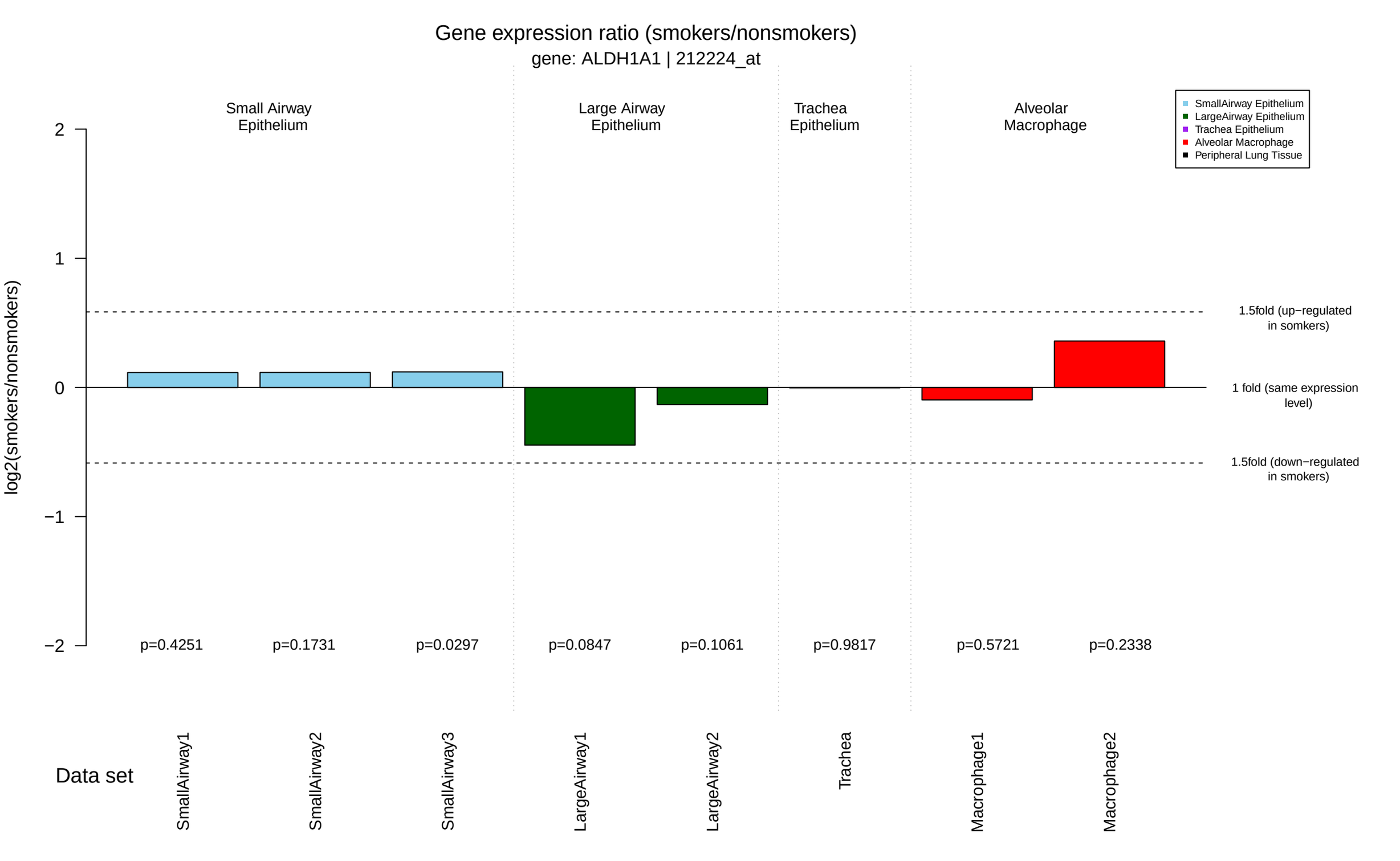
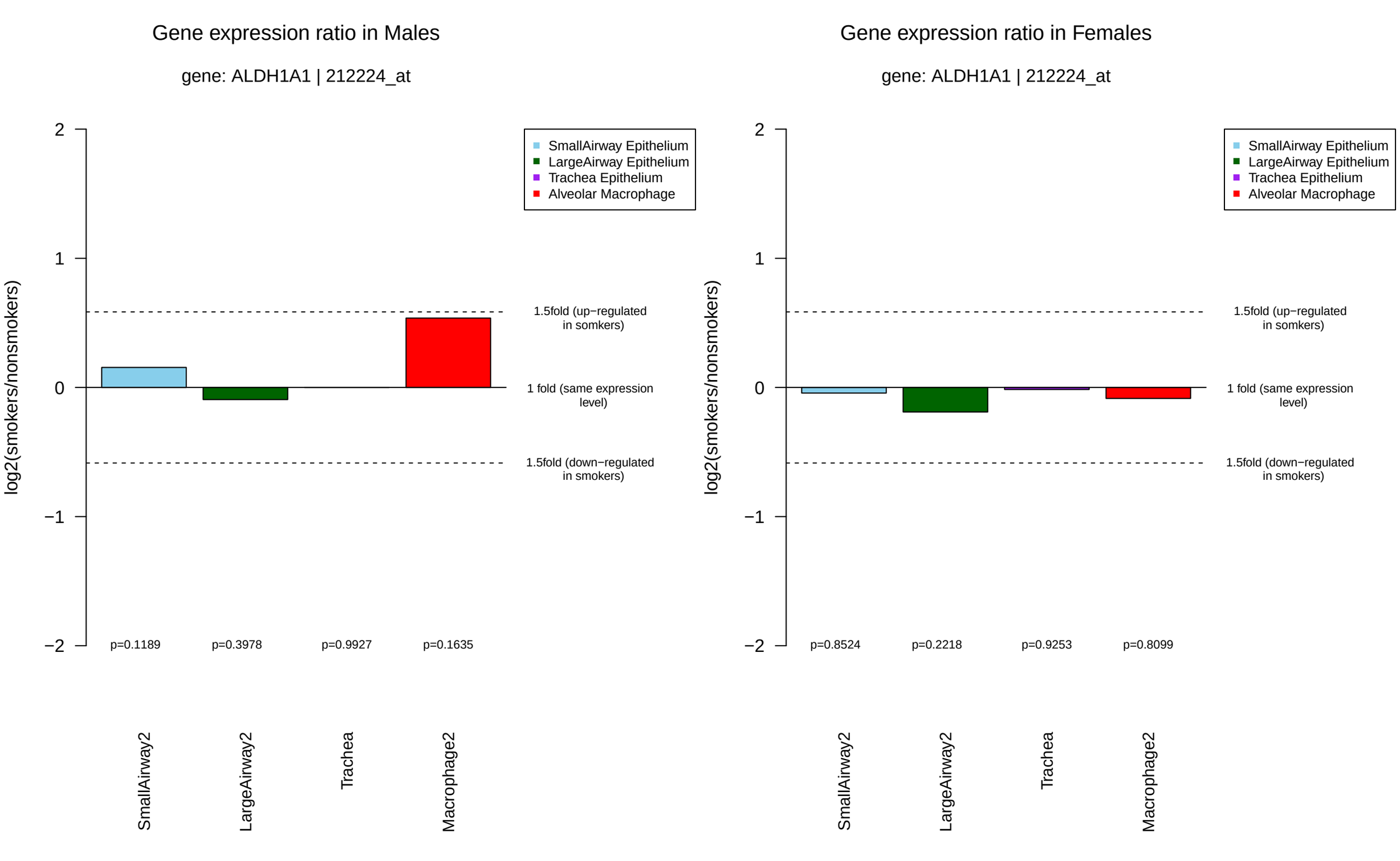
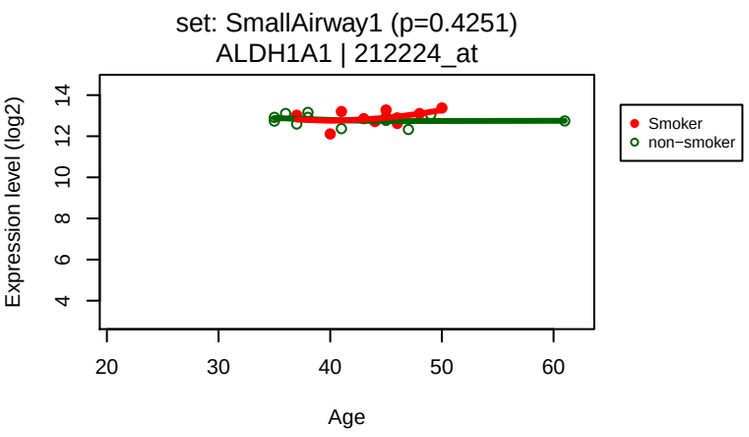
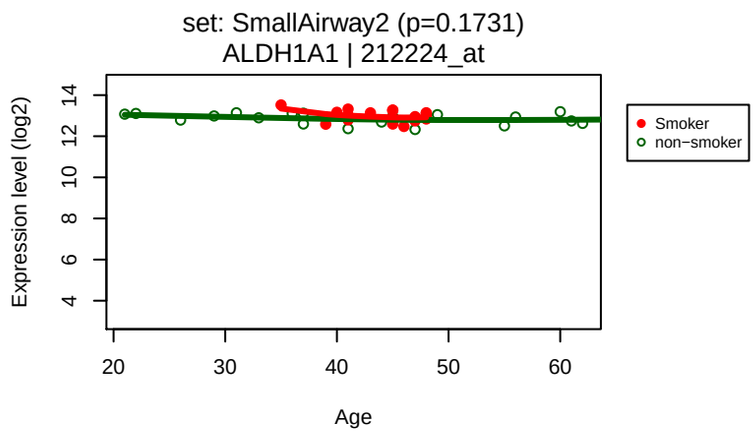
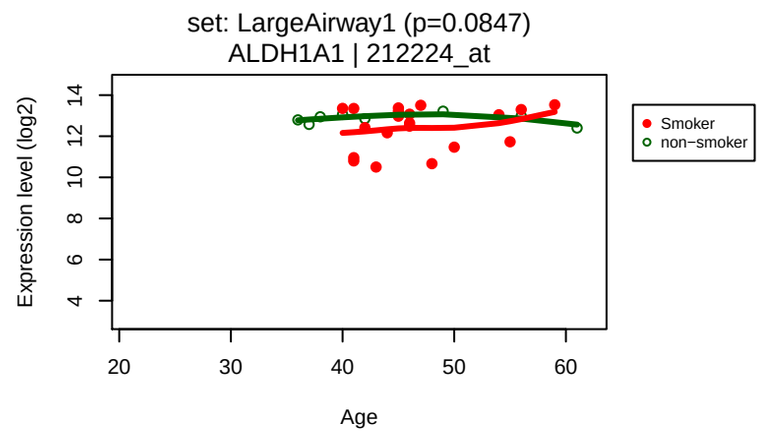

Gene expression of smokers and nonsmokers
gene: ALDH1A1 | 212224_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4251
p=0.1731
p=0.0297
p=0.0847
p=0.1061
p=0.9817
p=0.5721
p=0.2338
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

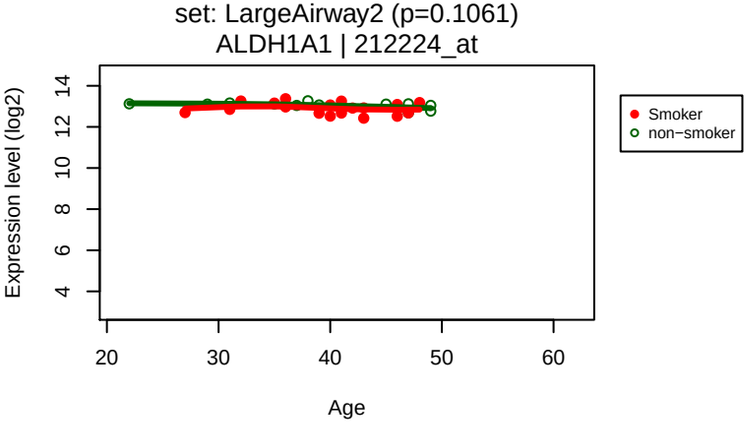

Gene expression ratio (smokers/nonsmokers)
gene: ALDH1A1 | 212224_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4251
p=0.1731
p=0.0297
p=0.0847
p=0.1061
p=0.9817
p=0.5721
p=0.2338
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

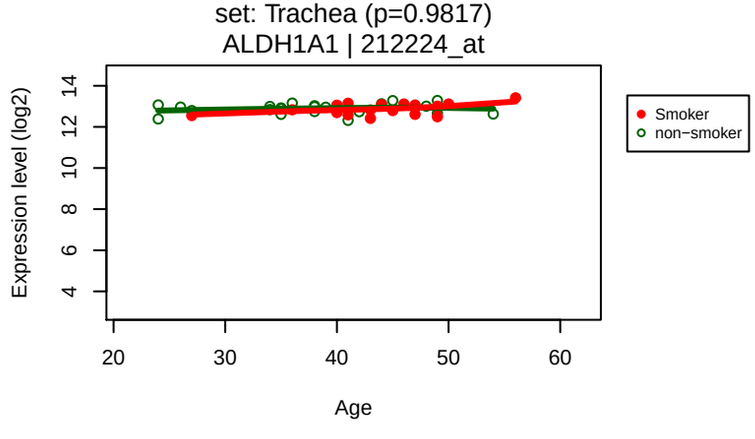

Gene expression ratio in Males
gene: ALDH1A1 | 212224_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1189
p=0.3978
p=0.9927
p=0.1635
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALDH1A1 | 212224_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8524
p=0.2218
p=0.9253
p=0.8099
SmallAirway2
LargeAirway2
Trachea
Macrophage2

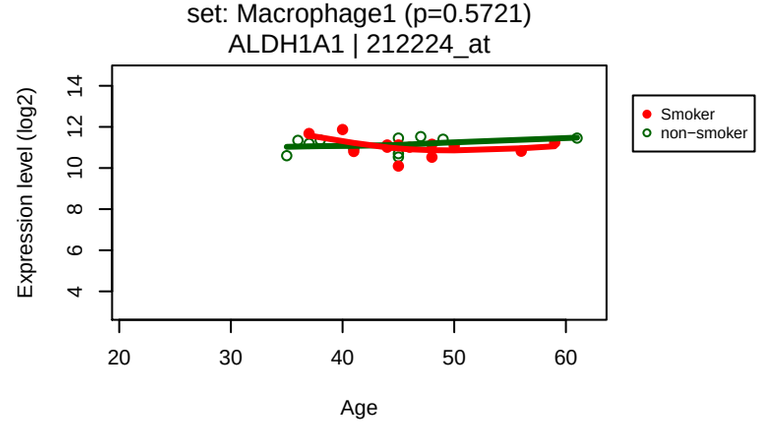

set: SmallAirway1 (p=0.4251)
ALDH1A1 | 212224_at
Expression level (log2)
Age
Smoker
non−smoker

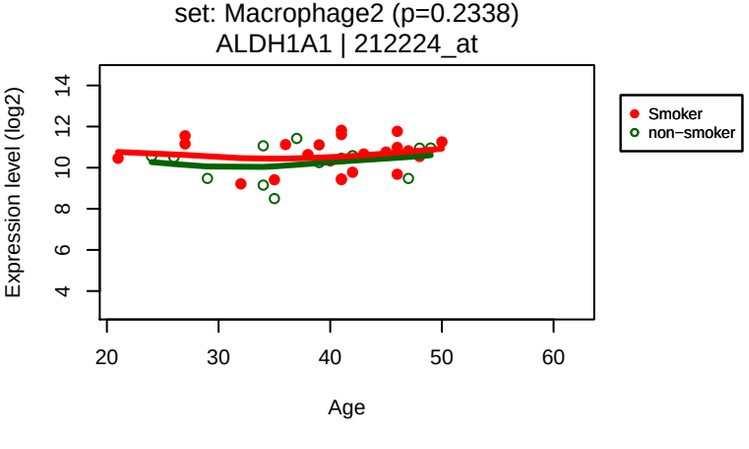

set: SmallAirway2 (p=0.1731)
ALDH1A1 | 212224_at
Expression level (log2)
Age
Smoker
non−smoker

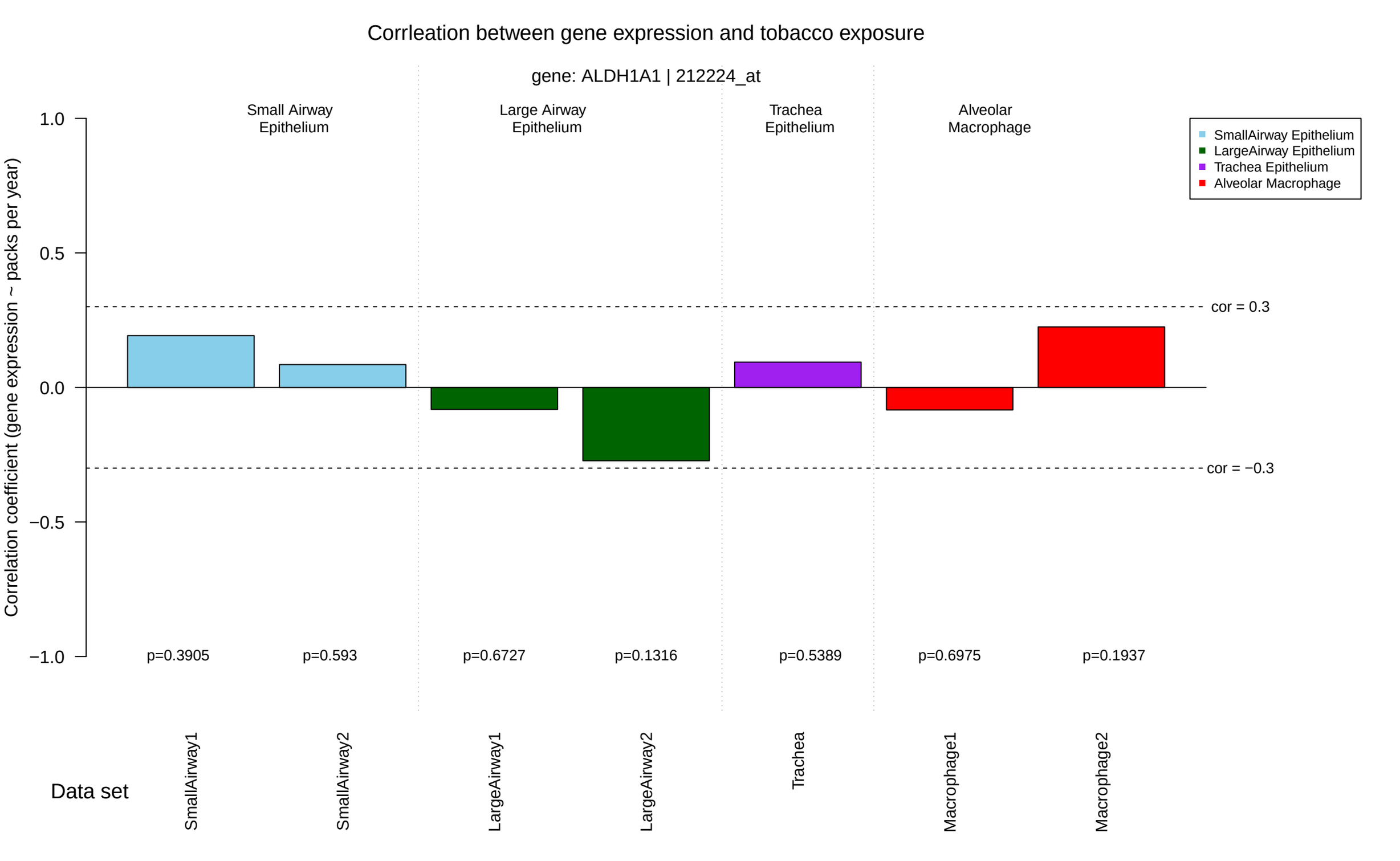

set: LargeAirway1 (p=0.0847)
ALDH1A1 | 212224_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1061)
ALDH1A1 | 212224_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9817)
ALDH1A1 | 212224_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5721)
ALDH1A1 | 212224_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2338)
ALDH1A1 | 212224_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALDH1A1 | 212224_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3905
p=0.593
p=0.6727
p=0.1316
p=0.5389
p=0.6975
p=0.1937
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2